

2.4 Creatine Kinase

▼ What is the main role of creatine kinase?

- To catalyse the phosphorylation of ADP to ATP via the breakdown of creatine phosphate.

▼ Describe the structure of a creatine kinase isoenzyme.

- It is a dimeric structure which consists of two subunits from M or B.

▼ What are the three main creatine kinase isoenzyme forms?

- MB, MM, BB

▼ Where are they found?

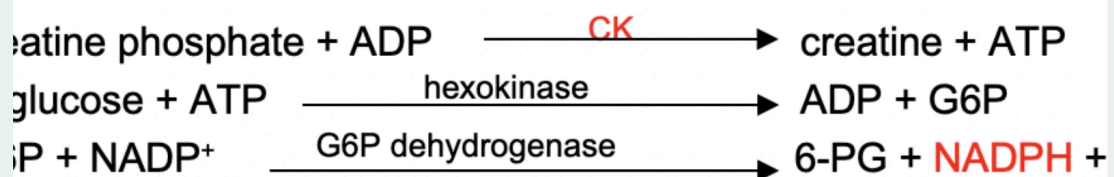
- **BM** - cardiac muscle
- **BB** - brain
- **MM** - skeletal muscle

▼ What is a special consideration regarding one of the dimeric isoenzymes?

- MB is only found in heart muscle cells.

▼ What is a coupled enzyme assay?

- It is a test where you **measure the activity of an enzyme**
- You do this by coupling the enzyme to another one, allowing a series of reactions to proceed and **measuring a product in one of these reactions to indirectly determine the initial enzyme's activity**



coupled assay to detect creatine kinase activity.

▼ How does measuring levels of different isoforms of creatine kinase help us?

- It can help us to understand where exactly muscle activity is taking place in the body (i.e in brain, muscle or heart muscle).

▼ What is the isoelectric point?

- The pH at which the charge on a molecule is neutral.

▼ What is a myocardial infarct, and what causes it?

- The **death of heart muscle cells**, caused by a lack of oxygen, often caused by **atherosclerosis**

▼ How does a lack of oxygen lead to cell death?

- A lack of oxygen means that **oxidative phosphorylation cannot take place** and so **ATP is not produced** in the final step of aerobic respiration
- This leads to **malfunction** of vital cellular components such as **protein pumps** which are **crucial to maintaining ion concentrations** across cell membranes (i.e sodium-potassium pump)
- This poor regulation can lead to cell death if ions are imbalanced and this leads to cells bursting too in a process called **lysis**
 - **Cell shrinkage** can also lead to **cell death**

▼ What can be measured as an indicator for cell death?

- Levels of certain intracellular proteins (i.e creatine kinase, lactate dehydrogenase)
- Abnormally high levels of these proteins outside cells could indicate high levels of **cell lysis**

▼ What assumption does the statement "the volume of cardiac muscle cell (myocyte) death can be measured by the amount of BM Creatine Kinase Isomer present in the serum" lie on?

- That **each myocyte releases an equal amount of creatine kinase** when it dies (independent of size)

▼ What laboratory technique can be used to identify different creatine kinase isoforms and why?

- **Gel electrophoresis**
- This is because they **separate subjects on the basis of charge or molecular weight**
- All three isoforms of creatine kinase have **different isoelectric points** and so will be **separated on the basis of charge**.

▼ What three markers can be used as an indication of myocardial infarction?

- Serum glutamate oxaloacetate transaminase (SGOT)
- lactate dehydrogenase (LDH)
- Cardiac troponin
- Obviously MB Creatine Kinase.

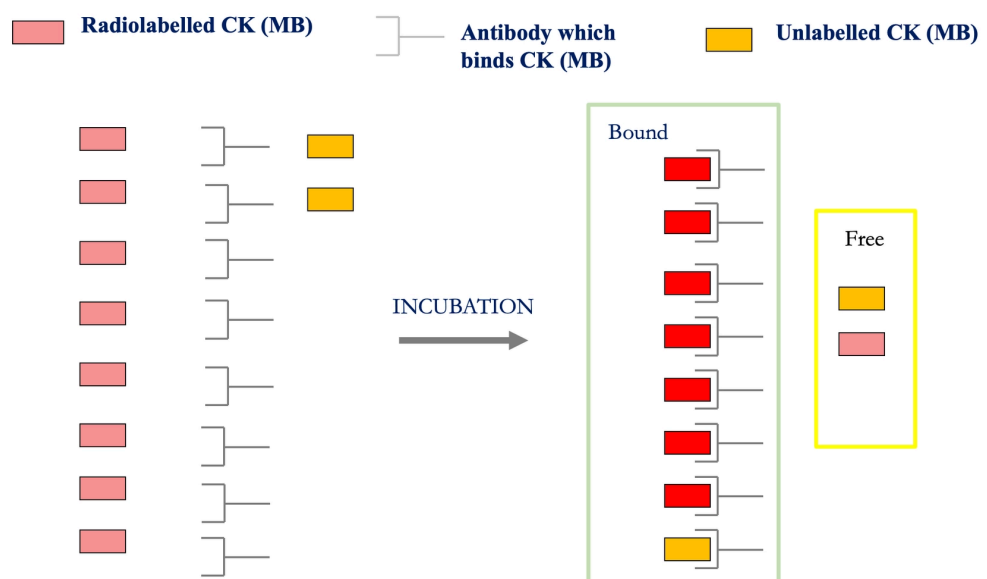
▼ What is an immunoassay?

- This is a molecular technique that utilises **antibodies to bind onto specific molecules**, that **change colour after binding** and can be **measured**

▼ Describe how an immunoassay can be used in measuring serum levels of creatine kinase (CK).

- A blood/serum sample can be obtained from the patient
- This **sample can be mixed with labelled CK that changes colour when bound to the antibodies** and it also has native antibodies
- The level of binding by the labelled CK can be **measured using the colour saturation produced by this CK binding to antibodies**
- Then, using a standard curve we can assess serum levels of CK by converting saturation to serum CK levels. The **lower the colour saturation, the higher the levels of native CK are**.
 - This is because **native CK is unlabelled**, hence if there is a **higher level of native CK**, there will be **less labelled CK able to bind to antibodies and change colour** due to **increased competition for the antibodies**

Presence small amount unlabelled CK (MB)



Presence large amount unlabelled CK (MB)

